Trabajo de Fin de Máster
Máster Universitario en Lógica, Computación e IA

A Study on Foundational Visual-Language Models for Histopathology Image Analysis

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Leandro Candau Sánchez de Ybargüen

Sevilla, 2026

Resumen

- **Contexto.** Clasificación de imágenes histopatológicas bajo el cuello de botella recurrente del dominio: datos etiquetados limitados y vocabularios de etiquetas estrechos.
- **Enfoque.** Aprendizaje contrastivo imagen–texto al estilo CLIP con codificadores fundacionales congelados. Sobre una línea base controlada (ResNet50 + BERT congelados) se evalúa una matriz de ablaciones de una sola variable, entrenadas en Kaggle LC25000 (5 clases de tejido pulmonar y colónico) y evaluadas tanto sobre LC25000 como sobre tres conjuntos externos no vistos en entrenamiento.
- **Resultado principal.** El *preentrenamiento del codificador de imagen* es, con diferencia, la palanca dominante. Manteniendo el resto del pipeline idéntico y sustituyendo el ResNet50 de propósito general (ImageNet) por el ViT-B/32 preentrenado en histopatología de PLIP, la macro-F1 externa mejora en $+0.067$ en colon de cercanía (NCT-CRC), $+0.027$ en colon de otro hospital (Chaoyang) y $+0.335$ en pulmón de otro órgano (LungHist700).
- **Descomposición con DINOv2.** Un ViT-B/14 con preentrenamiento autosupervisado en imágenes naturales generales (sin histopatología) eleva LungHist700 en $+0.152$ sobre la línea base — aproximadamente la mitad de la ganancia de PLIP. La otra mitad procede del preentrenamiento específico en histopatología.
- **Hallazgos secundarios.** De las otras cinco intervenciones probadas, la normalización de tinción Macenko (referencia NCT-NORM) es la única palanca del lado de la entrada que produce una ganancia positiva medible bajo desplazamiento ($+0.030$ en Chaoyang, $+0.020$ en LungHist700), pero su magnitud queda dominada por la palanca del codificador de imagen en los mismos conjuntos y regresa en NCT-CRC (-0.022, que llega ya normalizado por Macenko en su *release*). Los codificadores de texto biomédicos, el clasificador de entropía cruzada sin CLIP y las variaciones de formato de *prompt* no mejoran la generalización. La ventaja de CLIP sobre un clasificador no-CLIP aparece sólo bajo desplazamiento más fuerte, vía la geometría *coseno + L2-normalizada + escalada por temperatura* del clasificador.
- **Aporte metodológico.** El reporte por órgano es esencial: agregar los conjuntos externos en un único número promedio oculta el cambio de régimen entre colon (transferencia parcial con ResNet50/ImageNet congelado) y pulmón (fallo categórico del mismo *backbone*).

Abstract

- **Context.** Histopathology image classification under the domain's recurring bottleneck: limited labelled data and narrow label vocabularies.
- **Approach.** CLIP-style image–text contrastive learning with frozen foundation-model encoders. A single-variable ablation matrix is evaluated over a controlled baseline (frozen ResNet50 + frozen BERT), trained on Kaggle LC25000 (5 lung and colon tissue classes) and evaluated on both LC25000 and three additional external datasets the model has never seen.
- **Headline result.** The *image encoder's pretraining* is by far the dominant lever. Holding the rest of the pipeline identical and swapping the general-purpose ResNet50 (ImageNet) for PLIP's histopathology-pretrained ViT-B/32 lifts external macro-F1 by $+0.067$ on close-shift colon (NCT-CRC), $+0.027$ on different-hospital colon (Chaoyang), and $+0.335$ on different-source lung (LungHist700).
- **DINOv2 decomposition.** A ViT-B/14 with self-supervised pretraining on general natural images (no histopathology data) lifts LungHist700 by $+0.152$ over baseline — roughly half of PLIP's gain. The other half comes from histopathology-specific pretraining.
- **Secondary findings.** Among the other five interventions tested, Macenko stain normalization (NCT-NORM reference) is the only input-side lever that produces a measurable positive gain under shift ($+0.030$ on Chaoyang, $+0.020$ on LungHist700), but its magnitude is dominated by the image-encoder lever on the same datasets and it regresses on NCT-CRC (-0.022, which arrives Macenko-normalised at release). Biomedical text encoders, a non-CLIP cross-entropy classifier, and prompt-format variations do not improve generalization. CLIP's edge over a non-CLIP classifier appears only under heavier shift, via the *bounded-cosine* + *L2-normalised* + *temperature-scaled* classifier geometry.
- **Methodological contribution.** Per-organ external reporting is essential: aggregating external datasets into a single mean-OOD number hides the regime change between colon (partial transfer with a frozen ImageNet ResNet50) and lung (categorical failure of the same backbone).

Contents

1 Introduction

Learning directly from raw text about images is a promising alternative which leverages a much broader source of supervision.

RADFORD ET AL., 2021

1.1 Motivation

- Clinical relevance of histopathology classification.
- Limited labelled data — a recurring limitation in this domain.
- Foundation models [1] — able to do different tasks with little task-specific supervision; in the vision–language line, CLIP [2] is the paradigm under study here.
- Project question (refined during the project): *"Where does the pretraining of a CLIP-style histopathology pipeline pay off, and where does it not?"* — evaluated across head-side, loss-side, text-side, and image-side interventions, both in-distribution and on three external datasets covering two organs.
- Framing: CLIP as the object of study, not the classifier. Findings should generalise to other scientific domains where short or missing labels are the norm.

1.2 Project Scope Evolution

- Initial direction → semantic segmentation. Abandoned because annotated segmentation data was too scarce within the time budget.
- Pivot → image-level classification, same domain, reframed question.
- Training source: LC25000 [3] (5-class, lung + colon).
- External evaluation: NCT-CRC-HE-7K [4] (close-shift colon), Chaoyang [5] (different-hospital colon), and LungHist700 [6] (different-source lung).
- Reference points: PLIP [7] zero-shot (no LC25000 supervision) and a plain image classifier (no CLIP architecture).

1.3 Research Questions

- **RQ1** — What baseline classification and generalization behaviour does a frozen general-purpose CLIP pipeline (ImageNet ResNet50 + BERT) achieve on LC25000, and what failure modes appear?

- **RQ2** — Does the CLIP paradigm validate its *foundational-model* premise — learn one task, generalize to tasks it was not specifically trained for — relative to a non-CLIP classifier on the same image encoder?
- **RQ3** — Does stain normalization help when we evaluate on datasets the model was never trained on?
- **RQ4** — Does using a text encoder pretrained on biomedical text (BioBERT [8], PubMedBERT [9]), instead of generic BERT [10], improve performance under short class-name prompts?
- **RQ5** — Does the image encoder's *pretraining choice* matter? Specifically: replacing the general-purpose ResNet50 [11] with PLIP's histopathology-pretrained ViT-B/32 [7, 12], and with DINOv2's general-domain self-supervised ViT-B/14 [13] as a control that isolates "ViT architecture and self-supervision" from "histopathology pretraining specifically".
- **RQ6** — How well does each variant generalize to data the model has never seen, evaluated across three external datasets that span shift severity and two organs?

1.4 Contributions

The central contribution of this project is a controlled measurement of how much each major design choice in a CLIP-style histopathology pipeline actually matters, with every claim framed as a delta over the same baseline — frozen ResNet50 + frozen BERT, single fixed prompt, stratified 80/10/10 split on LC25000 — and iterated through one experiment at a time in Chapter 4.

Headline findings (full results in Chapter 4):

- **Image encoder pretraining is the dominant lever.** Replacing ResNet50 with PLIP's pretrained ViT-B/32 (histopathology image–text contrastive) gains $+0.067$ F1 on close-shift colon (NCT-CRC), $+0.027$ on different-hospital colon (Chaoyang), and $+0.335$ on different-source lung (LungHist700). No other intervention tested moves the needle comparably.
- **The lung-side advantage decomposes evenly.** The DINOv2 control (same general ViT architecture family as PLIP, self-supervised on general natural images, no histopathology data) lifts LungHist700 by $+0.152$ over baseline — about half of PLIP's $+0.335$ gap. The other half is attributable to histopathology pretraining specifically.
- **Five other interventions move the needle far less, or only in narrow regimes.** Stain Macenko (NCT-NORM reference) is the only input-side lever that produces a measurable positive gain under shift ($+0.030$ on Chaoyang, $+0.020$ on LungHist700), but its magnitude is dominated by the image-encoder lever on the same datasets and it regresses on NCT-CRC (-0.022, which arrives Macenko-normalised at release). Biomedical text encoders (BioBERT, PubMedBERT) tie or trail baseline across all 6 grid cells on NCT, and the best biomedical cell (PubMedBERT + name_only) extended to Chaoyang and LungHist700 confirms no transfer benefit on any of the three datasets. A non-CLIP plain classifier ties CLIP in-distribution and on close-shift external data but loses on heavier shift; prompt-format variations within the text encoder ablation are within noise.
- **CLIP's foundational-model premise is validated under shift, not in-distribution.** CLIP and a plain CE classifier tie at saturation in-distribution and on close-shift external; CLIP's edge appears under heavier shift via the bounded-cosine + temperature-scaled classifier geometry, which doesn't degenerate the way high-confidence softmax does.

Methodological contributions:

- Single-variable-change protocol: each experiment differs from the baseline in exactly one design choice, so the resulting delta is attributable.
- Per-organ external reporting: results are presented per dataset, not averaged; aggregate mean-OOD numbers would hide the cross-organ regime change observed in this study.
- Reproducibility: artifact bundle (fixed-seed split, cached DINOv2 features, weights, metric files, plots) versioned alongside the report in the project repository.
- Explicit written discussion of dataset and methodological limits (Chapter 5).

1.5 Document Structure

- Chapter 1 (Introduction) — motivation, scope, research questions, contributions.
- Chapter 2 (State of the Art) — foundation models, multimodal contrastive lineage, pathology FMs, where this thesis fits.
- Chapter 3 (Methodology) — CLIP architecture, image and text encoders, datasets, preprocessing, baseline, evaluation protocol, reproducibility.
- Chapter 4 (Results and Discussion) — the experiments and what each measured, ending in a synthesis section.
- Chapter 5 (Conclusions and Future Work) — one-paragraph-per-RQ conclusions, limitations, future work.

2 State of the Art

AI is undergoing a paradigm shift with the rise of models [...] that are trained on broad data at scale and are adaptable to a wide range of downstream tasks.

Bommasani et al., 2021

- Foundation models framing (Bommasani et al. [1], 2021).
- CLIP and multimodal contrastive learning (Radford et al. [2], 2021; see also ALIGN [14]) — the paradigm under study.
- Self-supervised visual pretraining — contrastive (SimCLR [15], MoCo [16]), self-distillation (DINO [17]), and scaled SSL on diverse natural images (DINOv2 [13]).
- Pathology shift: per-tile supervised CNNs $\rightarrow$ foundation models (2016–2024).
- Pathology-specific foundation models:
  - TransPath [18] (2022) — transformer + SSL.
  - Phikon [19] (2023) — masked image modelling.
  - PLIP [7] (2023) — CLIP on medical Twitter.
  - CONCH [20] (2024) — larger pathology VLM.
  - UNI [21] (2024), BiomedCLIP [22] (2023), MUSK [23] (2025) — recent entrants.
- Open challenges: benchmarks, dataset bias, cross-institution generalization, compute scale.
- Where this project fits: small-scale empirical study of general-purpose foundation backbones on pathology classification, with single-variable ablations.

3 Methodology

3.1 Overview

- One-page chapter map + project question recap.
- Reading order: background → baseline → ablations.

3.2 Histopathology Digital Imaging

- H&E staining.
- Whole-Slide Imaging + patch extraction.
- Stain variability and normalization (Macenko [24], Reinhard [25], Vahadane [26]).
- Why the domain is hard for general-purpose VLMs: domain gap from natural images, no per-image captions in clinical archives, dataset artifacts that can drive performance.

3.3 Contrastive Vision-Language Learning (CLIP)

3.3.1 Why CLIP?

- CLIP as object of study, not the classifier.
- Capabilities CLIP provides over a softmax head:
  - Open-vocabulary classification (new classes via prompts proven by adding new samples).
  - Semantic similarity output (cosine vs natural language).
  - Prompt engineering as a sensitivity lever.

3.3.2 Architecture and training signal

- Dual-encoder, shared d-dimensional embedding space ($d = 256$ throughout this thesis).
- Symmetric InfoNCE loss with a learnable temperature τ (formal definition below).
- In-batch false-negative regime at small batch + few classes (discussed under *Caveats* below).
- Prompt-based zero-shot classification at inference: an image is classified by argmax of cosine similarity against the encoded class prompts.
- CLIP-paper sample-level retrieval → doesn't apply to our setup: every image in a class shares the same prompt, so retrieval would measure class assignment rather than sample identification.

InfoNCE loss. For a batch of N matched image–text pairs $\{(I_i, T_i)\}_{i=1}^{N}$, the model embeds each side into the shared d-dimensional space and produces two L2-normalised batches of embeddings, $\{\mathbf{v}_i\}$ and $\{\mathbf{t}_i\}$. Define the scaled cosine similarity between image i and prompt j as

$$s_{ij} \;=\; \frac{\mathbf{v}_i \cdot \mathbf{t}_j}{\tau}, \tag{3.1}$$

where τ is the learnable temperature introduced below. For each image i the image-to-text contrastive loss treats the matched prompt i as the positive and the other $N-1$ prompts in the same batch as negatives; the row-wise softmax cross-entropy then reads

$$\mathscr{L}_{I\to T} \;=\; -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(s_{ii})}{\sum_{j=1}^{N}\exp(s_{ij})}. \tag{3.2}$$

The symmetric InfoNCE loss [**?**, 2] averages this with the analogous column-wise text-to-image loss $\mathscr{L}_{T\to I}$ (in which each prompt is matched against its image positive and the $N-1$ batch images as negatives):

$$\mathscr{L}_{\text{InfoNCE}} \;=\; \tfrac{1}{2}\left(\mathscr{L}_{I\to T} + \mathscr{L}_{T\to I}\right). \tag{3.3}$$

Geometrically, minimising $\mathscr{L}_{\text{InfoNCE}}$ pulls matched (image, prompt) pairs together in cosine-similarity space and pushes the $N-1$ batch-internal mismatches apart.

Learnable temperature. The scalar τ in equation (3.1) controls the sharpness of the softmax in equation (3.2): a smaller τ peaks the distribution and pushes the model toward sharply confident positives, while a larger τ smooths it. Following CLIP [2] we store $\log(1/\tau)$ as the trainable variable (which guarantees $\tau > 0$ throughout training without an explicit constraint) and initialise it to $\log(1/0.07) \approx 2.66$, the CLIP-paper default. Gradient descent moves it freely during training. Empirically $\log(1/\tau)$ drifts upward (i.e. τ shrinks) as the embedding space stabilises, reflecting the model's preference for sharper positives as the contrastive task becomes easier; the converged value is reported alongside the loss curves in Chapter 4.

Caveats. The InfoNCE objective above assumes the $N-1$ in-batch negatives are *genuine* mismatches. With our 5-class LC25000 supervision and batch size $N{=}32$ (Table 3.2), the expected number of true mismatches per image is $0.8 \times (N-1) \approx 24.8$ but the expected number of *false negatives* (other batch entries sharing the image's class, and therefore its prompt) is ~ 6. The loss therefore penalises some semantically-correct alignments. We do not mitigate this in the present study — it is documented as a limitation in Chapter 5.

Figure 3.1 reproduces the canonical CLIP paradigm as introduced by Radford et al. [2]: contrastive image–text pretraining (panel 1) builds a shared embedding space; class prompts then become the zero-shot classifier (panels 2–3). Figure 3.2 is the specific instantiation of that paradigm used in this thesis — short histopathology class prompts, 256-d projection heads on top of frozen foundation-model encoders, and a single fixed prompt strategy used at both training and evaluation time.

3.3.3 PLIP: a histopathology-pretrained CLIP-style model

- Huang et al. [7] (2023) — a CLIP-style model pretrained on histopathology image–text pairs.
- Training corpus: OpenPath, $\sim$208k histopathology image–text pairs scraped from Twitter conversations among pathologists, supplemented with LAION subset filtering.
- Architecture: CLIP ViT-B/32 vision encoder paired with a CLIP-style text encoder, both trained contrastively from a CLIP initialisation.
- Used in this thesis in two distinct ways:
  - As a *zero-shot reference baseline* (§3.6): the full PLIP model (image + text encoders) used off-the-shelf for cosine-similarity classification on all four datasets (LC25000 + the three external evaluation sets).
  - As a *pretrained image encoder* in our image-encoder ablation (§3.4.2): PLIP's ViT-B/32 vision encoder paired with our trained projection head and a separate BERT text encoder.

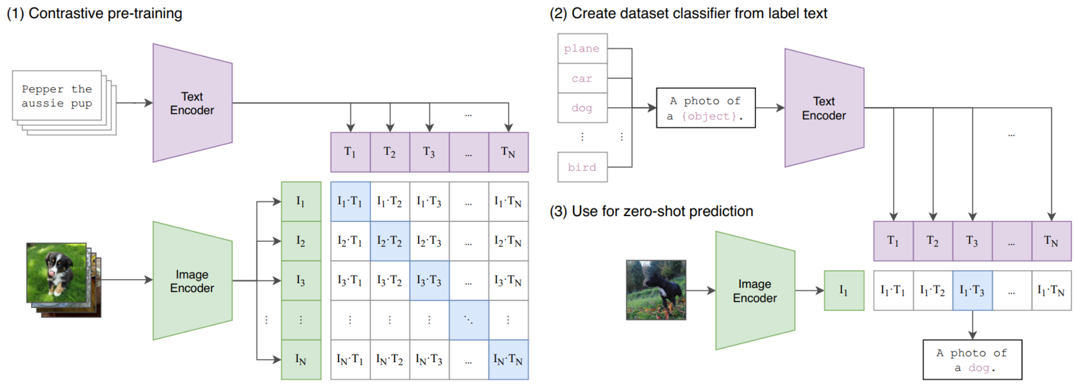


Figure 3.1 The CLIP paradigm as introduced by Radford et al., 2021 (Figure 1 of the original paper, reproduced for illustration). **(1) Contrastive pre-training**: two encoders project a batch of image–text pairs into a shared embedding space; the in-batch cosine-similarity matrix is supervised so the diagonal pairs (matched) score higher than off-diagonals (mismatched). **(2) Build a dataset-specific classifier from label text**: any new task is cast as text by encoding class-name prompts through the same text encoder. **(3) Zero-shot prediction**: an unseen image is classified by cosine-similarity argmax against the class prompts — no task-specific training needed..

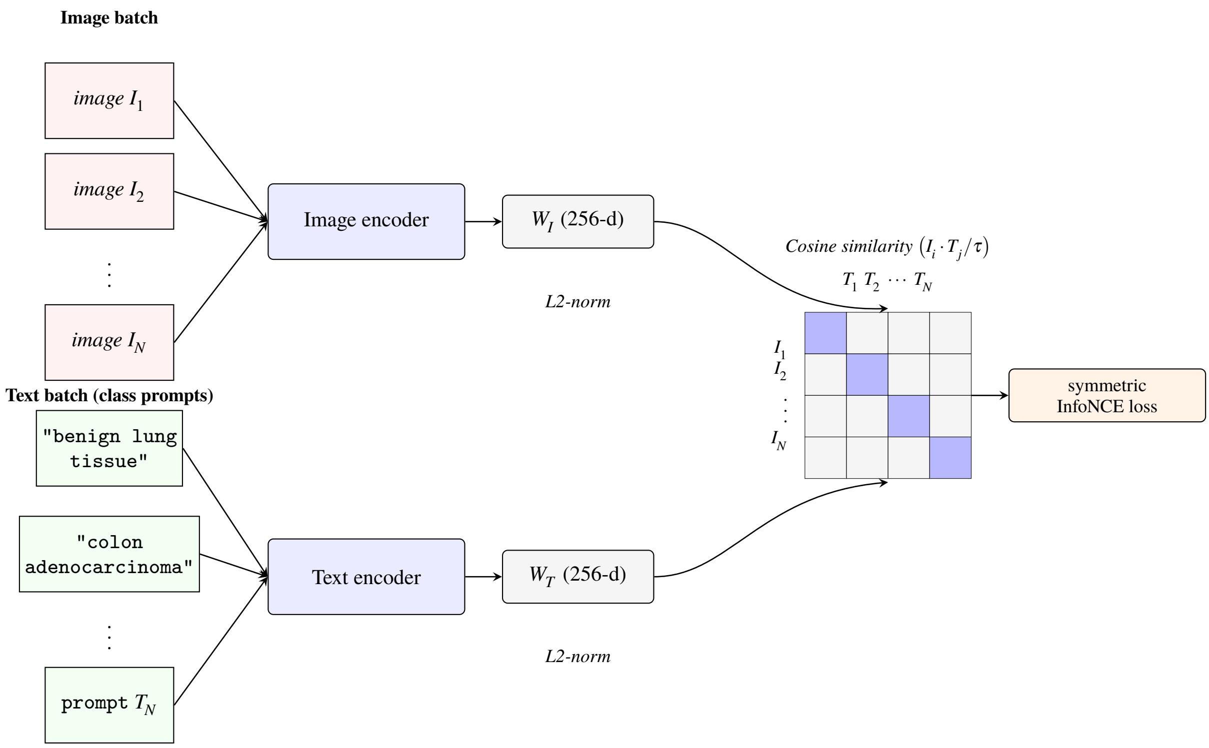


Figure 3.2 CLIP dual-encoder architecture used in this project. Two independent encoders project a batch of N image–text pairs into a shared d-dimensional embedding space (here $d = 256$, the projection head's output). After L2-normalisation and a learnable temperature τ, the $N \times N$ cosine-similarity matrix between every image and every prompt is the input to the symmetric InfoNCE loss. The diagonal (blue) entries are the positive pairs; off-diagonal entries are negatives within the batch..

3.4 Image Encoders

3.4.1 ResNet50 (CNN, ImageNet supervised)

- He et al. [11] (2016), residual learning.
- $\sim$23M parameters, 2048-d output via global average pooling.
- Supervised pretraining on ImageNet-1K ($\sim$1.3M natural images, 1000 classes).
- Role in this thesis: general-purpose image-encoder baseline; carries the assumption that ImageNet features transfer broadly.

3.4.2 Vision Transformer (ViT)

- Dosovitskiy et al. [12] (2020).
- Patch-based architecture: image split into $N \times N$ patches, each linearly embedded as a token, then processed by a stack of transformer encoder layers with self-attention.
- The "B" sizing (ViT-Base): 12 layers, hidden size 768.
- The "/16", "/32", "/14" suffixes denote patch size; smaller patch = more tokens = finer spatial granularity at the same input resolution.
- Role in this thesis: we use **PLIP's pretrained ViT-B/32** as one image-encoder variant. PLIP itself is described in §3.3.3; this section covers only the ViT architecture used by PLIP's vision encoder.

3.4.3 DINOv2 (self-supervised on general natural images)

- Oquab et al. [13] (2024).
- Self-supervised pretraining on LVD-142M ($\sim$142M curated general natural images); no labels, no text supervision, no histopathology data.
- Self-distillation between a teacher and student network on augmented crops of the same image.
- We use the `base` size with patch 14 (**ViT-B/14**).
- Role in this thesis: a control. Same general ViT architecture family as PLIP's vision encoder, similar self-supervised paradigm, but no histopathology pretraining — isolates "ViT + self-supervision" from "histopathology pretraining specifically" in the image-encoder ablation.

3.5 Text Encoders

- Transformer [27] + BERT [10] — text encoder used by baseline.
- BioBERT [8] / PubMedBERT [9] — biomedical text encoder swap ablation.

3.6 Reference baselines: PLIP zero-shot and a plain image classifier

3.6.1 PLIP zero-shot

- Uses the full PLIP model (§3.3.3) — both image and text encoders, frozen — for zero-shot classification.
- Class prediction: cosine similarity between the image embedding and the five PLIP-encoded LC25000 class-prompt embeddings, argmax.
- No LC25000 supervision; no projection head trained on our data.

- Evaluated on all four datasets: LC25000 (training source) plus the three external evaluation datasets (NCT-CRC, Chaoyang, LungHist700; the latter two are queued follow-up runs at the time of writing).
- Role: isolates the value of histopathology pretraining alone, without any of our fine-tuning.

3.6.2 Plain image classifier

- Drops the CLIP paradigm entirely: no text encoder, no projection heads, no learnable temperature, no contrastive loss.
- Architecture: ResNet50 → Global Average Pooling → BatchNorm → Dense(5) trained with sparse cross-entropy.
- Role: isolates what the CLIP paradigm itself contributes once the image encoder, data and adaptation budget are held constant.

3.7 Training Dataset: LC25000

- Borkowski et al. [3] (2019).
- 5 tissue classes (lung, colon).
- 25 000 patches from 1 250 originals via rotation/flip → no further geometric augmentation.
- Known limits: single institution, possible stain artifacts, no patient-level grouping.

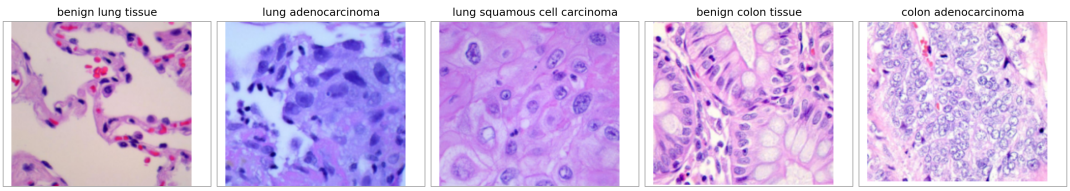


Figure 3.3 Example image per class from LC25000. Left-to-right: benign lung tissue, lung adenocarcinoma, lung squamous cell carcinoma, benign colon tissue, colon adenocarcinoma. The two benign classes appear visually distinct from the cancer classes; within the cancers, lung adenocarcinoma vs lung squamous cell carcinoma is the hardest discrimination in our experiments (and a clinical confound)..

Table 3.1 LC25000 class counts and stratified split sizes..

Class	**Total**	**Train**	**Val**	**Test**
benign lung tissue	5 000	4 000	500	500
lung adenocarcinoma	5 000	4 000	500	500
lung squamous cell carcinoma	5 000	4 000	500	500
benign colon tissue	5 000	4 000	500	500
colon adenocarcinoma	5 000	4 000	500	500
Total	25 000	20 000	2 500	2 500

3.8 External Evaluation Datasets

- Role: data the model has never seen during training. We evaluate every LC25000-trained variant against three external datasets, chosen to span shift severity and two organs.
- Order presented below corresponds to increasing shift severity relative to LC25000.

3.8.1 NCT-CRC-HE-7K

- Kather et al. [4] (2019), Zenodo record 1214456.
- 7 180 patches across 9 tissue classes; we restrict to NORM ($\leftrightarrow$ benign colon tissue) and TUM ($\leftrightarrow$ colon adenocarcinoma) for 1 974 patches scored (741 NORM + 1 233 TUM).
- 224×224 patches at $\sim 20\times$ magnification, Macenko-normalised at release.
- Role: close-shift colon evaluation (stylistically similar acquisition to LC25000).

3.8.2 Chaoyang

- Zhu et al. [5] (2022; preprint 2021).
- 6 160 patches across 4 classes (normal, serrated, adenocarcinoma, adenoma); we restrict to normal ($\leftrightarrow$ benign colon, $n = 1816$) and adenocarcinoma ($\leftrightarrow$ colon adeno, $n = 2244$) for 4 060 patches scored.
- 512×512 patches at $20\times$, Beijing Chaoyang Hospital.
- Documented label noise in the training split; the test split is labelled by consensus of three pathologists.
- Role: different-hospital colon evaluation (different scanner, different staining, label noise).

3.8.3 LungHist700

- Tabatabaei et al. [6] (2024, *Scientific Data*).
- $\sim$691 high-resolution images at $20\times$ and $40\times$ covering normal lung tissue plus adenocarcinoma and squamous-cell carcinoma at three differentiation grades each; we restrict to the $20\times$ subset for 3-class scoring (359 images: 85 normal + 146 AdC + 128 SqC).
- Role: different-source lung evaluation; the only multi-class scoring among the external datasets.

3.9 Preprocessing and Input Pipeline

3.9.1 Image preprocessing (per-backbone)

- ResNet50: RGB / 255.0 (i.e., $[0, 1]$ linear scaling).
- ViT-B/32 (PLIP): CLIP-standard mean/std normalisation
  mean = $[0.481, 0.458, 0.408]$, std = $[0.269, 0.261, 0.276]$.
- ViT-B/14 (DINOv2): ImageNet mean/std normalisation
  mean = $[0.485, 0.456, 0.406]$, std = $[0.229, 0.224, 0.225]$.
- All inputs resized to 224×224 bilinear. No additional augmentation (LC25000 already includes rotation/flip augmentation at release).
- Each backbone expects the input distribution it was pretrained on; mismatched preprocessing produces degraded features — see §3.9.4.

3.9.2 Text preprocessing

- All text encoders use BERT-family WordPiece tokenizers (BERT, BioBERT, PubMedBERT, and PLIP's text encoder).
- Fixed sequence length 24 tokens (covers all class-prompt formats).
- Composed prompt format: `"A histopathology image of {class_name}."`.

3.9.3 LC25000-specific cleanup

- 1 280 byte-identical duplicate files identified by MD5 grouping; excluded via `src/data/lc25000_-dedupe_exclusions.json`.
- Stratified 80/10/10 split, seed 42, persisted to canonical JSON.
- The canonical split file predates the dedupe pass, so the 1 280 duplicates still cross train/test for the runs reported; flagged as a limitation in Chapter 5.

3.9.4 Methodological note on preprocessing

- Mid-project we caught that the pipeline was applying Caffe-style BGR preprocessing to a backbone expecting $[0,1]$ RGB — a mismatch from two libraries shipping different ResNet50 checkpoints under the same name.
- The error was surfaced via a linear-probe diagnostic on the frozen features (the probe macro-F1 was well above the trained CLIP ceiling), and fixed with a centralised image loader at `src/data/images.py`.
- Beyond the project-specific bug, the takeaway is that linear-probing a frozen feature extractor against the trained model's ceiling is a cheap sanity check worth running as a project-foundation step.

3.10 The Baseline

- Frozen ResNet50 + frozen BERT.
- Projection heads (Dense + LayerNorm) $\rightarrow$ 256-d shared space.
- Learnable temperature, CLIP-style init.
- Symmetric InfoNCE, AdamW, grad clipping.
- Single fixed prompt at train + eval.
- Float32 throughout.

Table 3.2 Baseline hyperparameters. Ablations change exactly one row..

Hyperparameter	Value
Image encoder	ResNet50 (ImageNet preset), frozen
Text encoder	BERT base uncased, frozen
Projection head	Dense(256, no bias) $\rightarrow$ LayerNorm
Embedding dimension	256
Image size	224×224
Text sequence length	24 WordPiece tokens
Batch size	32
Initial temperature	$\tau = 0.07$
Loss	symmetric InfoNCE
Optimizer	AdamW
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
Gradient clipnorm	1.0
Max epochs	30
EarlyStopping patience	5
ReduceLROnPlateau	factor 0.5, patience 2, min 10^{-6}
Numerical precision	float32
Random seed	42

3.11 Evaluation Protocol

3.11.1 Metrics and visualizations

- Classification rule (CLIP variants): argmax of cosine similarity between the image embedding and the five LC25000 class-prompt embeddings.
- Classification rule (plain classifier): argmax of softmax over Dense(5).
- Global metrics: accuracy, balanced accuracy, macro-F1, weighted F1.
- Per-class: precision, recall, F1, support.
- Confusion matrices: raw counts + row-normalised by true class.
- Visualizations: shared image + prompt UMAP (training source), and cross-distribution UMAP grid per external dataset (LC25000 dots + external triangles overlaid in one fit).

3.11.2 External-dataset evaluation: restricted-argmax protocol

- Each variant exposes all 5 LC25000 class prompts at inference, but on external evaluation we argmax *only* over the LC25000 classes that map to the external dataset's organ:
  - NCT and Chaoyang (colon datasets): argmax restricted to {benign colon tissue, colon adenocarcinoma}.
  - LungHist700 (lung dataset): argmax restricted to {benign lung tissue, lung adenocarcinoma, lung squamous cell carcinoma}.
- Why: external datasets cover only a subset of LC25000 classes; an unrestricted argmax would routinely predict the model's strongest out-of-organ class on these images and produce uninformative misclassifications.
- Predictions into a non-target LC25000 class count as wrong but are reported separately when informative for confusion analysis.
- Pure inference: every variant reuses its LC25000-trained checkpoint; no fine-tuning on external sets.
- Class-mapping rules per dataset are detailed in §3.8.

3.11.3 Per-organ reporting

- Results are reported per external dataset, not averaged across datasets.
- Aggregating across datasets would hide the cross-organ regime change observed in this study (Chapter 4); a single mean-OOD number collapses an order-of-magnitude difference between organs into one misleading point.
- Per-dataset reporting is therefore an evaluation choice with explicit methodological motivation.

3.12 Experimental Design and Ablation Protocol

- Single-variable-change principle.
- Matrix grouped by axis of variation; full matrix in Table 4.1 of Chapter 4.
- Scope boundaries: no patient-level grouping, no human-expert evaluation, no architecture-controlled comparison vs CONCH [20] / BiomedCLIP [22] / MUSK [23]. Deferred to future work.

3.13 Computational Setup and Reproducibility

- Hardware: Google Colab T4 (occasional L4 for longer runs); local CPU/GPU fallback for development and analysis.
- Main stack: TensorFlow + KerasHub for the CLIP-style training pipeline and all downstream evaluation.
- Auxiliary stack: PyTorch + HuggingFace transformers for image-encoder feature extraction when the encoder lacks a TensorFlow port (DINOv2; see below).
- Library versions pinned in `requirements.txt`.
- Reproducibility: seed 42 across NumPy, TensorFlow, and PyTorch; `PYTHONHASHSEED` set; `tf.config.experimental.enable_op_determinism()`. Runs are bit-exact on the same hardware; small numerical drift can appear across GPU types but does not change the classification outcomes reported.
- DINOv2 has no TensorFlow port in our environment; its image features are precomputed once in PyTorch and cached to disk as HDF5 (one file per dataset, keyed by image relative path). The TensorFlow training and evaluation pipelines then read from cache. Mathematically equivalent to running the frozen backbone online, since the backbone is frozen and each image always produces the same feature vector. Total cache footprint $\sim$100 MB across LC25000 plus the three external datasets.

4 Results and Discussion

4.1 Experimental Overview

4.1.1 Research questions

The chapter is built around six research questions, each targeting a single design axis of a CLIP-style histopathology pipeline: the baseline reference, the CLIP-versus-non-CLIP choice, the input-side preprocessing, the text encoder, the image encoder, and cross-dataset generalisation. Five of them (RQ1–RQ5) are answered by a dedicated experiment section in this chapter; RQ6 is cross-cutting and is addressed by the external evaluations run alongside every other experiment. The experiment matrix in Table 4.1 below maps each RQ to the controlled comparison that addresses it.

- **RQ1** — What baseline classification and generalisation behaviour does a frozen general-purpose CLIP pipeline achieve on LC25000, and what failure modes appear?
- **RQ2** — Does the CLIP paradigm validate its *foundational-model* premise — learning one task and generalising to tasks it was not specifically trained for — relative to a non-CLIP classifier on the same image encoder?
- **RQ3** — Does stain normalisation, an input-side preprocessing that targets a known shortcut in histopathology, help when evaluating on datasets the model was never trained on?
- **RQ4** — Does using a text encoder pretrained on biomedical text (BioBERT, PubMedBERT), instead of generic BERT, improve performance under short class-name prompts?
- **RQ5** — Does the image encoder's *pretraining choice* matter? We test two alternatives to the ImageNet-supervised ResNet50 baseline: (a) a Visual Transformer pretrained on histopathology image–text pairs (PLIP), and (b) a Visual Transformer pretrained self-supervised on general natural images (DINOv2), which serves as a control to separate the contribution of "ViT architecture and self-supervision" from "histopathology pretraining specifically".
- **RQ6** — How well does each variant generalise to data the model has never seen? We evaluate every variant against three external datasets across two organs that span shift severity: close-shift colon, different-hospital colon, and different-source lung.

4.1.2 Training source and external datasets

Every variant in this chapter is trained on a single source dataset and evaluated on three external datasets it has never seen during training. The training source is LC25000 [3], already introduced in §4.2.1: every trained variant is fit on it, and the in-distribution test split yields the macro-F1 we refer to as *training-source results* throughout the chapter. The three external datasets — the source of every *generalisation* number in the chapter — are listed in order of shift severity:

- **NCT-CRC-HE-7K** [4], close-shift colon: 1 974 scored patches over the `NORM` and `TUM` classes, mapping to *benign colon tissue* and *colon adenocarcinoma* in LC25000.

Table 4.1 Experiment matrix. Each experiment isolates exactly one axis of variation relative to the baseline. Sections §4.2–§4.7 follow this order, ending with image-encoder pretraining (the chapter's headline finding)..

Experiment	Image encoder	Text encoder	Loss	Preprocessing	Axis tested
Baseline (§4.2)	ResNet50 (ImageNet)	BERT base	InfoNCE	RGB / 255	reference
Stain normalization (§4.3)	ResNet50 (ImageNet)	BERT base	InfoNCE	Macenko / Reinhard / Vahadane	input-side
PLIP zero-shot (§4.4)	ViT-B/32 (PLIP / OpenPath)	PLIP text encoder	none (zero-shot)	CLIP mean/std	reference: no LC25000 supervision
Foundational vs plain (§4.5)	ResNet50 (ImageNet)	— (no text)	cross-entropy	RGB / 255	architecture: CLIP paradigm vs plain classifier
Text encoder (§4.6)	ResNet50 (ImageNet)	BioBERT / PubMedBERT	InfoNCE	RGB / 255	text encoder pretraining
Image encoder (§4.7)	ViT-B/32 (PLIP); ViT-B/14 (DINOv2)	BERT base	InfoNCE	varies	image encoder pretraining

- **Chaoyang** [5], different-hospital colon: 4 060 scored patches (`normal` and `adenocarcinoma`; the `serrated` and `adenoma` classes have no clean LC25000 analogue and are excluded from scoring), mapping to the same two LC25000 colon classes as NCT.
- **LungHist700** [6], different-source lung: 359 images at 20× magnification (the 40× subset is excluded), mapping to LC25000's *benign lung tissue*, *lung adenocarcinoma*, and *lung squamous cell carcinoma*.

Full dataset provenance, acquisition details, and per-dataset preprocessing live in Methodology (§3.8); the bullets above keep only what is needed to interpret the external-evaluation tables in the rest of the chapter.

Three protocol choices apply to every external evaluation that follows. First, **restricted argmax**: each variant exposes all 5 LC25000 class prompts at inference, but the argmax is restricted to the LC25000 classes that map to the external dataset's organ (2 colon classes for NCT and Chaoyang, 3 lung classes for LungHist700). Predictions into a non-target LC25000 class count as wrong but are reported separately when informative for confusion analysis. Second, **pure inference**: every variant reuses its LC25000-trained checkpoint without further adaptation; no fine-tuning is performed on the external datasets. Third, **per-dataset reporting**: results are presented per dataset rather than averaged. The motivation for that last choice is methodological — aggregating across datasets would average a colon-side partial transfer ($\sim$ 0.79–0.87 F1) with a lung-side categorical failure (0.305 F1 in the baseline), producing a single mean-OOD number that hides the cross-organ regime change that ends up driving the chapter's headline result in §4.7.

4.1.3 Metrics and visualizations

Each variant in this chapter is reported with the same set of metrics and visualisations. The choices and what each is meant to surface:

- **Macro-F1** is the headline metric. It averages the per-class F1 scores equally, which is the right behaviour for the per-class precision–recall trade-offs that drive the chapter's substantive comparisons. Formal definitions are in Methodology (§3.11).
- **Accuracy, balanced accuracy, weighted F1, and per-class precision/recall/F1** are reported alongside macro-F1 for completeness, but the chapter's interpretive weight is on macro-F1 and the per-class numbers.
- **Confusion matrices** (raw counts and row-normalised by true class) are reported per variant and per dataset — the most direct view of how a model distributes its errors across the available classes.
- **Training-source UMAP** projects LC25000 image embeddings and class-prompt embeddings into the same 2-D fit (cosine metric), letting the reader see cluster separation and the alignment between image clusters and prompt anchors.
- **Cross-distribution UMAP grid** (per external dataset, per variant) overlays LC25000 dots and external-dataset triangles in a single 2-D fit, asking whether the external dataset co-locates with the matching LC25000 class cluster or sits in its own region of feature space.

4.2 Baseline: frozen ResNet50 and frozen BERT

4.2.1 Configuration

The baseline pipeline pairs a frozen ResNet50 image encoder, taken from the standard ImageNet-supervised checkpoint, with a frozen BERT-base-uncased text encoder; only a pair of small projection heads — a single 256-wide `Dense` layer with no bias, followed by `LayerNorm` — and a scalar temperature τ are trained on top. This is deliberately the most general-purpose CLIP one can assemble. ResNet50 is the canonical CNN baseline of the past decade and BERT-base is the canonical text encoder; neither has been exposed to histology during pretraining, and neither has been tuned for the task. The reason to start here is mechanical: every subsequent variant in this chapter changes exactly one piece of this configuration, and the resulting delta is only interpretable if the reference is something recognisable rather than something already adapted to the data.

Everything else is held constant across the variants that follow. The training data is the LC25000 dataset, with a 80/10/10 train–validation–test split at random seed 42; a single fixed composed prompt, ‘‘`A histopathology image of {class_name}.`’’, is used at both training and inference; the loss is symmetric InfoNCE; the optimiser is AdamW with learning rate 1×10^{-3} and weight decay 1×10^{-4}; and training runs for at most 30 epochs, with EarlyStopping (patience 5, restore best weights) terminating it earlier on every run. The remaining design choices — batch size, image and sequence dimensions, initial temperature, gradient clipping, learning-rate schedule, numerical precision, weight initialisation — are not referenced again in this chapter and are gathered in the baseline hyperparameter table in Methodology (Table 3.2).

4.2.2 Training behaviour and in-distribution results

On the LC25000 test split, the baseline reaches a macro-F1 of 0.996. The model has only two trainable projection heads and a scalar temperature τ on top of frozen encoders — around 720,000 parameters in total — so a saturated in-distribution score would invite a memorisation reading without a check on the training dynamics. Figure 4.1 provides that check: the training and validation InfoNCE losses (equation (3.3)) descend together over the 30-epoch budget and stay close at convergence. The learnable temperature τ (equation (3.1)), initialised at the CLIP-paper value of 0.07, *rises* during training, peaks at ≈ 0.090 around epoch 10, and settles near 0.087. This is the opposite direction from the CLIP paper's $\tau \to 0.01$ behaviour on uniquely-paired image–text data, and is consistent with our shared-prompt setup: when multiple images per batch share the same class prompt (the false-negative caveat in §3.3), softening τ reduces the penalty those false negatives impose on otherwise-correct alignments.

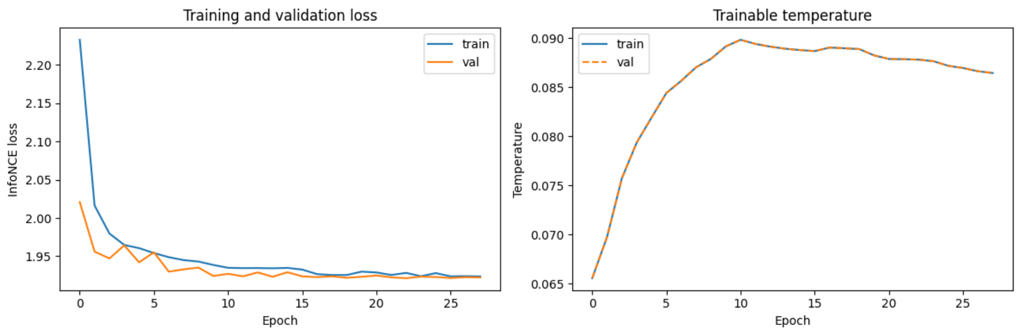
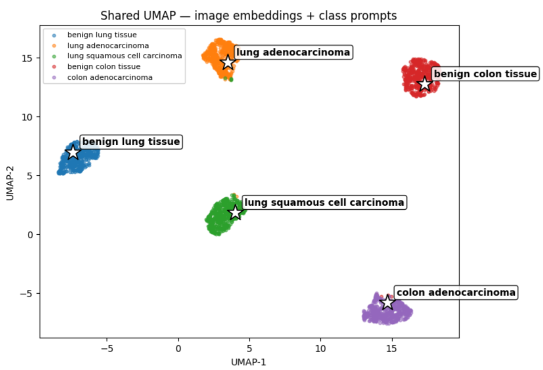
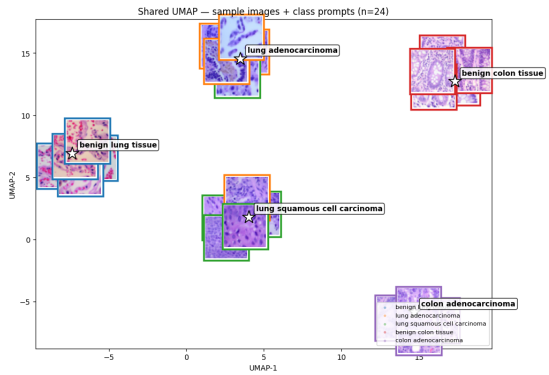


Figure 4.1 Baseline training (blue) and validation (orange) InfoNCE loss across epochs (left), and the learnable temperature τ on the same horizontal axis (right)..

The embedding space the model converges to is shown in Figure 4.2. Each LC25000 class forms a distinct cluster of image embeddings and the five class-prompt anchors (encoded through frozen BERT and projected to the same 256-d space) sit at the centre of their corresponding clusters — the expected post-InfoNCE alignment between text and image, with the trained projection heads doing the work that the frozen text encoder alone could not. The lung-adenocarcinoma cluster sits closest to lung squamous cell carcinoma in the 2D projection, foreshadowing the off-diagonal confusion-matrix mass discussed below.



(a) Vanilla scatter.



(b) Sample-image thumbnails.

Figure 4.2 UMAP of the baseline embedding space on the LC25000 test split. Two views of the same fit: (a) per-class scatter with the five class-prompt anchors marked, (b) the same fit overlaid with sample-image thumbnails..

The confusion matrix (Figure 4.3) confirms that the saturated 0.996 is genuinely close to ceiling: the benign tissues and lung squamous cell carcinoma separate cleanly, with the off-diagonal mass concentrated on lung adenocarcinoma — the hardest discrimination in the LC25000 schema. In-distribution, thus, does not discriminate the baseline from any of the trained variants studied in subsequent sections (all of them saturate at ~ 0.99); the meaningful comparison is the external evaluation that follows in §4.2.3.

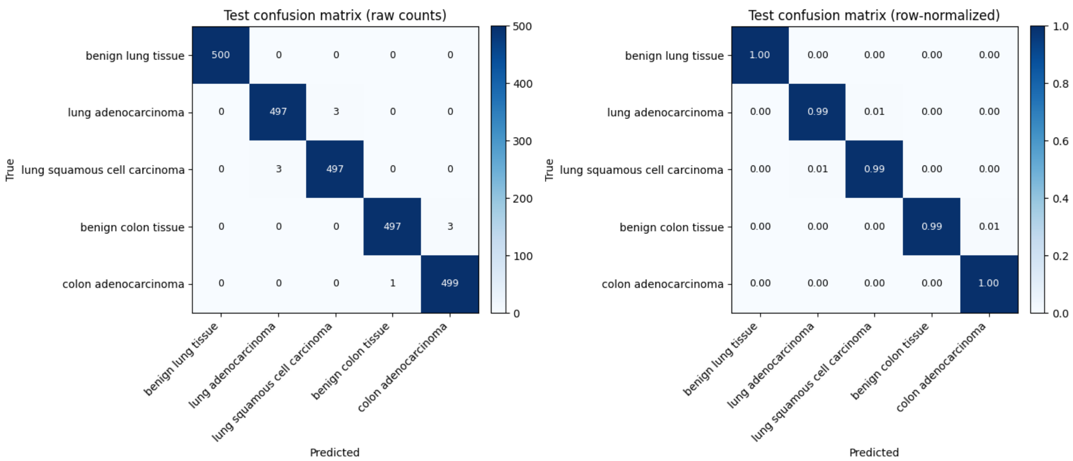


Figure 4.3 Baseline confusion matrix on the LC25000 test split. Left: raw counts. Right: row-normalised by true class..

4.2.3 Generalization to external datasets

The baseline's macro-F1 on each external dataset, restricted-argmax over the target-organ LC25000 classes, is reported in Table 4.2. The result splits in two. On the two colon datasets the baseline transfers partially — 0.866 on NCT-CRC-HE-7K, the dataset stylistically closest to LC25000's colon distribution, and 0.786 on Chaoyang, where a different hospital, scanner, and staining protocol drag the score down by ~ 0.08 F1 even though the organ is unchanged. On lung the picture is qualitatively different: **0.305 macro-F1 on LungHist700's 3-class problem, where chance is 0.33**. The colon-versus-lung gap dwarfs every within-organ shift in the table — the central observation the rest of the chapter sets out to explain.

Table 4.2 Baseline generalization to the three external datasets, restricted-argmax over the target-organ LC25000 classes..

External dataset (organ)	Macro-F1	Accuracy	Bal. acc.	Weighted F1
NCT-CRC-HE-7K (colon)	**0.866**	0.870	0.866	0.871
Chaoyang (colon)	**0.786**	0.796	0.782	0.791
LungHist700 (lung, 20×)	**0.305**	0.390	0.383	0.301

The lung results are not a noisy degradation of the colon transfer; they are a different failure mode. Per-class recall on LungHist700 breaks down as 0.89 for lung squamous cell carcinoma, 0.07 for lung adenocarcinoma, and 0.19 for benign lung tissue. Back-calculated from the precision figures, **the model assigns "squamous" to roughly 84% of LungHist700 inputs regardless of true class** — and given that squamous accounts for only 36% of the ground truth, the near-chance macro-F1 is the direct consequence of that bias. Figure 4.4 shows the matching feature-space signature: all 359 LungHist700 samples collapse into a single tight cluster with no visible internal class structure. The frozen ImageNet-supervised ResNet50, never exposed to histology during pretraining, extracts no usable per-class variance from LungHist700 tissue, and the trained projection head has nothing to align with the three lung class prompts.

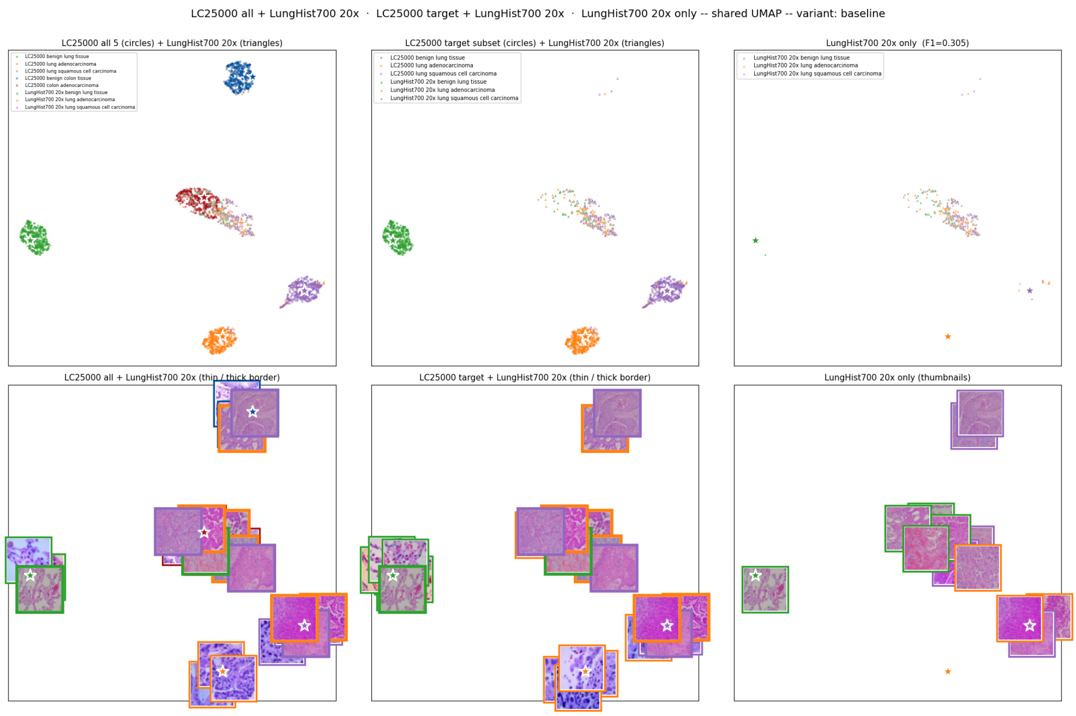


Figure 4.4 Baseline cross-distribution UMAP on LungHist700. LC25000 image embeddings (dots) and LungHist700 image embeddings (triangles) projected into the same 2-D fit..

The Chaoyang dataset sits between these two regimes (Figure 4.5). The two colon classes are partially separated in the projected space, and the Chaoyang triangles cluster in a region adjacent to — but not co-located with — the matching LC25000 colon clusters: enough signal for partial transfer, with a measurable cost from the cross-hospital distribution gap. Together with NCT and LungHist700, this exposes two distinct generalisation regimes for the experiments that follow. There is a colon regime, where frozen ImageNet features carry good-enough structure for partial transfer, and a lung regime, where they carry no usable structure at all. The central question the rest of this chapter is built around: is the lung-side failure about the encoder's architecture, about its supervised pretraining on natural images, or about the absence of histopathology data in pretraining? The DINOv2 control (general-domain self-supervised pretraining, no histology) and the PLIP backbone (ViT pretrained on histopathology image–text pairs) jointly decompose that question, and the headline answer arrives in §4.7.

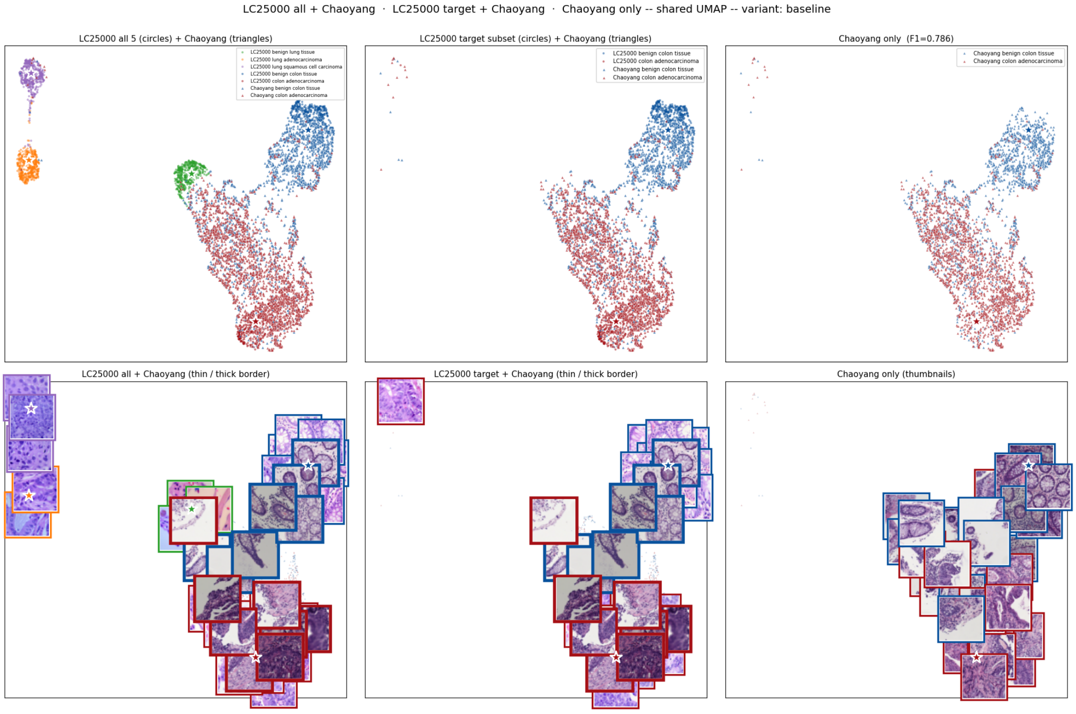


Figure 4.5 Baseline cross-distribution UMAP on Chaoyang. LC25000 image embeddings (dots) and Chaoyang image embeddings (triangles) projected into the same 2-D fit..

4.3 Stain normalization

4.3.1 Configuration

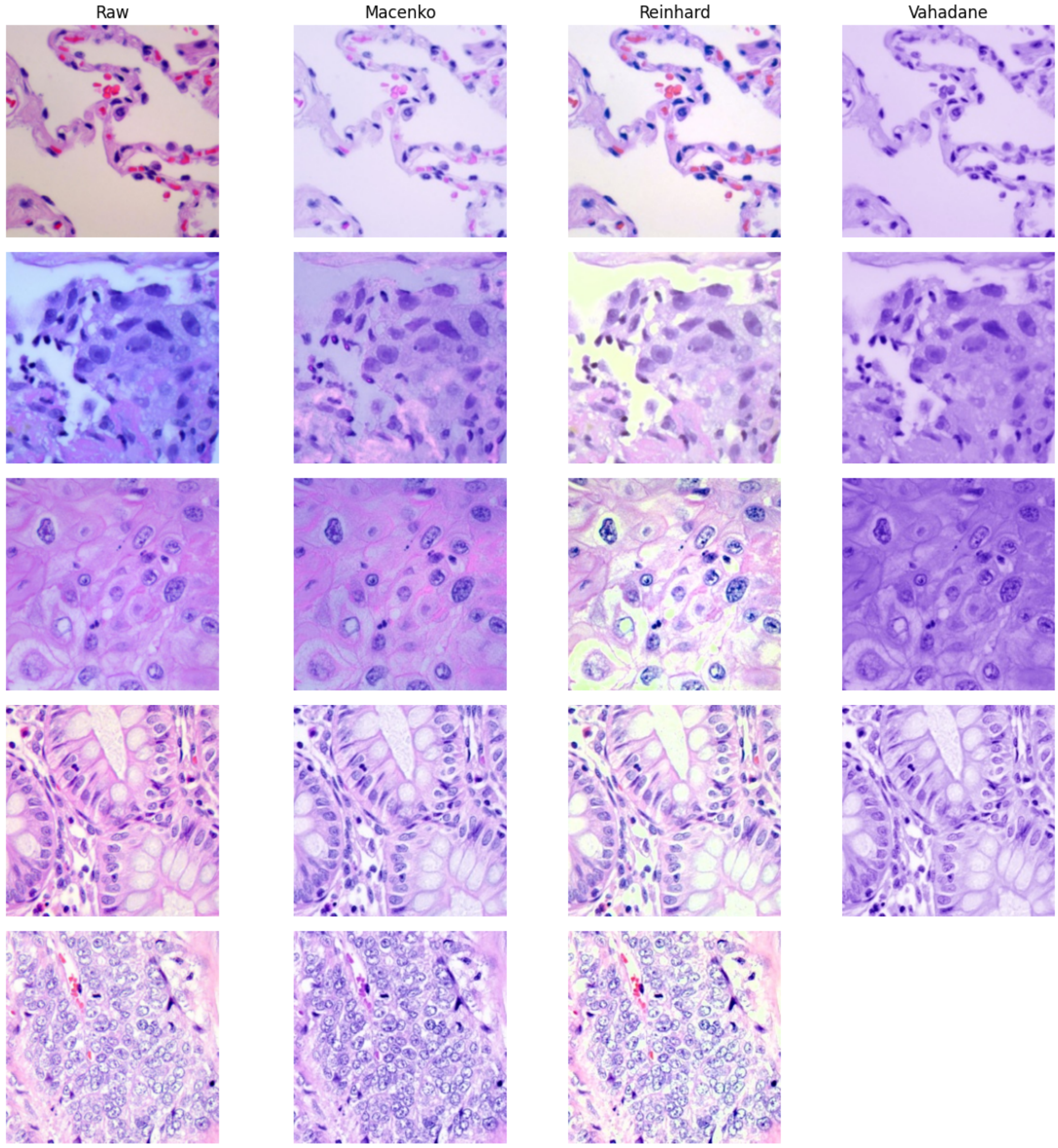
The stain-normalisation experiment keeps the baseline pipeline architecture identical to §4.2.1 and changes only the image-input distribution. Every LC25000 training image is Macenko-normalised [24] against a single fixed reference patch: the first `NORM` tile from NCT-CRC-HE-7K, abbreviated "NCT-NORM" below.

The choice of reference matters operationally. NCT-CRC is distributed already Macenko-normalised against an unpublished Kather-team reference [4]; by Macenko-normalising LC25000 onto an NCT-CRC tile, we land the training distribution in approximately the colour space NCT-CRC arrives in at release, which lets NCT-CRC be fed raw at evaluation and avoids the double-Macenko regime that historically zeroed out the OOD F1 in an earlier version of this experiment. Chaoyang and LungHist700, by contrast, arrive in their native un-normalised palettes, so for those two datasets the same Macenko(NCT-NORM) transform is applied at inference to bring them into the model's training-stain space. An alternative reference drawn from Chaoyang or LungHist700 — the latter being the most stain-distinct of the three external datasets — would have re-introduced the double-Macenko problem on NCT-CRC and required a separate retraining run for each reference tested; the NCT-NORM reference is therefore picked for methodological consistency rather than per-dataset optimisation, and reference rotation as an ablation axis is documented as a limitation in Chapter 5.

Reinhard and Vahadane were also explored as alternative normalisations in an exploratory pass but did not advance to the full external-evaluation ablation. Vahadane's NMF solver fails to converge on a fraction of LC25000 patches — visible in Figure 4.6(d), where the bottom-row `colon_aca` cell re-renders near-blank. Reinhard is a Lab-space colour-statistics transfer rather than a stain-decomposition method, included in the figure for visual comparison only.

4.3.2 Generalization to external datasets

On LC25000 the Macenko-trained checkpoint saturates at 0.986 macro-F1, alongside the baseline and every other trained variant in this study — in-distribution does not discriminate this lever, so the substantive



(a) Raw. (b) Macenko. (c) Reinhard. (d) Vahadane.

Figure 4.6 Stain normalization, one example per class. Column (a) shows the raw inputs (identical across methods, shown once for compactness); columns (b)–(d) show the same inputs after Macenko, Reinhard, and Vahadane normalization respectively. The bottom-row colon_aca cell in column (d) shows the Vahadane NMF solver failing to converge and re-rendering near-blank, an instability discussed in the interpretation..

comparison is the external evaluation reported in Table 4.3.

Table 4.3 Macenko(NCT-NORM) vs baseline on the three external evaluation datasets..

External dataset	Baseline F1	Macenko F1	Delta
NCT-CRC-HE-7K (close-shift colon)	0.866	0.844	−0.022
Chaoyang (different-hospital colon)	0.786	0.816	+0.030
LungHist700 (different-source lung, 20×)	0.305	0.325	+0.020

The lever's direction is data-dependent. On the two datasets that arrive in their native stain palette —

Chaoyang (+0.030 F1) and LungHist700 (+0.020 F1), both receiving Macenko at inference — normalisation moves the input closer to the model's training distribution and the frozen ImageNet ResNet50 produces better features. On NCT-CRC the picture inverts: NCT was already in approximately the target stain space at release, so training-time Macenko introduced a non-linear shift the frozen backbone could not absorb, and the eval-time pass-through cannot recover the small cost paid (−0.022 F1).

The magnitude is modest, even where the lever helps. The +0.020–+0.030 F1 gains on Chaoyang and LungHist700 are real but small; by comparison, swapping the image encoder to PLIP (§4.7) gains +0.027 on Chaoyang and **+0.335 on LungHist700 — an order of magnitude larger**. At the frozen-backbone scale evaluated here, stain normalisation is a useful but not dominant lever, and the protocol lesson worth carrying forward is the asymmetric application: Macenko is applied at inference only on inputs that arrive in their native stain palette, since doubling on a pre-normalised target is what historically broke the OOD F1 in this experiment.

4.4 PLIP zero-shot reference

4.4.1 Configuration

- Visual Transformer (ViT-B/32) with PLIP's OpenPath pretraining, paired with PLIP's matching text encoder. Both frozen.
- Evaluated zero-shot — no LC25000 supervision, no projection head trained on our data. Uses PLIP's own image and text encoders straight from the model release.
- Class prediction = cosine similarity to the five LC25000 class-prompt embeddings produced by PLIP's text encoder, argmax.
- Evaluated on the same training-source test split and the same external datasets as every other variant in this chapter.
- **Role in the chapter**: a reference point. PLIP zero-shot isolates the value of pathology pretraining alone — without any of our LC25000 fine-tuning — and shows what an off-the-shelf domain-pretrained model gives you on these tasks.

4.4.2 Training source results

- Macro-F1 on LC25000: **0.700**.
- Pathology pretraining alone (without any LC25000 supervision) clears the saturation null that every *trained* variant reaches at ~ 0.99 by ~ 0.30 F1 in the other direction — but it is the only variant in the chapter where in-distribution numbers are informative, because PLIP zero-shot has no LC25000 supervision to drive the saturation.
- Per-class gains concentrate on benign tissues; cancer subtypes are essentially tied with baseline. The improvement is image-side: PLIP produces visibly better class clusters (Figure 4.7) but the class-prompt embeddings still collapse, same short-prompt degeneracy as the baseline.

4.4.3 Generalization to external datasets

Table 4.4 PLIP zero-shot generalization across the three external datasets. No LC25000 supervision; class prediction by cosine similarity to PLIP-encoded prompts..

Dataset (organ)	Macro-F1	Accuracy	Bal. acc.	Weighted F1
NCT-CRC-HE-7K (colon)	0.661	0.557	0.610	0.679
Chaoyang (colon)	**0.731**	0.737	0.761	0.727
LungHist700 (lung, 20×)	0.544	0.621	0.567	0.583

(a) PLIP zero-shot.

(b) Baseline (for comparison).

Figure 4.7 Training-source UMAP comparison. PLIP zero-shot (a) shows visibly tighter, more separated class clusters than the baseline (b) without any LC25000 supervision — the gain is image-side..

- **Chaoyang > NCT.** PLIP zero-shot scores higher on Chaoyang (0.731) than on NCT-CRC (0.661), opposite of the pattern for every LC25000-trained variant (where NCT is the easier external dataset). Plausible reason: NCT-CRC is Macenko-normalised at release, pushing it away from the natural-staining distribution PLIP saw in OpenPath; Chaoyang's natural-stain images are stylistically closer to PLIP's pretraining corpus.
- **LungHist700 = 0.544.** The interesting cell. PLIP zero-shot has no LC25000 lung supervision yet outperforms the LC25000-trained ResNet50 variants (0.305) by +0.239 F1 on lung. Per-class recall: benign 0.16 / AdC 0.69 / SqC 0.84 — biased toward squamous like the ResNet50 variants, but not nearly as collapsed.
- Smallest training-source → external drop overall (−0.04 on NCT) — the "pretrained generalist" pattern: never the best, never the worst, but consistent across distributions.

4.4.4 What we learned

- Pathology pretraining produces a *meaningfully better image representation* (visible directly in the UMAP) even without LC25000 supervision.
- PLIP's off-the-shelf prompt geometry is weak for our specific 5-class LC25000 vocabulary — this decouples "the image representation is useful" from "the off-the-shelf classifier works".
- Section 4.7 returns to this distinction by keeping PLIP's image representation but replacing its prompt geometry with a projection head trained on LC25000.

4.5 Foundational model vs plain classifier

4.5.1 Configuration

- Same frozen ResNet50 image encoder as the baseline.
- Drops the CLIP paradigm entirely: no text encoder, no projection heads, no learnable temperature, no contrastive loss.
- Replaces it with a non-CLIP closed-set classifier: ResNet50 → global average pooling → BatchNorm → Dense(5) trained with sparse cross-entropy.
- Isolates what the CLIP paradigm itself contributes once the image encoder, data, and adaptation budget are held constant. Tests RQ2 (the foundational-model premise).

4.5.2 Training source results

- Macro-F1 on LC25000: **0.6369** (vs baseline 0.6273) *(both numbers pre-bug-fix; post-fix both saturate at* ~ 0.99*, indistinguishable in-distribution)*.
- In-distribution the two architectures tie within noise. The choice between contrastive (CLIP) and direct supervised (plain classifier) is invisible when the image features are fixed and the training distribution is the evaluation distribution.

4.5.3 Generalization to external datasets

Table 4.5 CLIP (baseline) vs plain classifier on the same frozen ResNet50 image encoder, macro-F1 across the training-source test split and three external datasets..

Variant	LC25000	NCT	Chaoyang	LungHist700
Baseline (CLIP InfoNCE)	0.996	0.866	**0.786**	**0.305**
Plain classifier (CE)	0.995	**0.880**	0.722	0.281
Δ (CLIP − plain)	−0.001	−0.014	+0.064	+0.024

- **Training source**: tied at ~ 0.99 (saturation null).
- **NCT-CRC (closest external dataset)**: plain CE marginally ahead by $+0.014$ F1.
- **Chaoyang (harder external)**: CLIP ahead by $+0.064$ F1 — plain CE drops to last.
- **LungHist700 (hardest external)**: both variants collapse near 3-class chance; CLIP retains a marginal edge.

Per-class signature on Chaoyang. Plain CE shows benign precision 0.96, benign recall 0.47 (high precision, low recall — the classifier is rarely predicting benign but is right when it does); adenocarcinoma precision 0.70, recall 0.98. The plain classifier is almost-degenerate "predict adenocarcinoma" under shift. The baseline CLIP variant is imperfect but balanced (benign 0.86/0.65, adeno 0.76/0.92), avoiding the collapse.

4.5.4 What we learned

- CLIP and plain CE are *interchangeable on easy distributions*; the choice of architecture is invisible when the model is evaluated on its training distribution.
- CLIP's edge appears *under shift severity and grows with it*: -0.014 F1 on close-shift NCT (plain ahead), $+0.064$ on different-hospital Chaoyang (CLIP ahead), $+0.024$ on different-source LungHist700.
- **Mechanism**: same image encoder, same image features. The difference is the classifier geometry. The plain CE softmax learns class *direction vectors* in feature space and is trained to maximise the logit margin; on shifted data those vectors don't move but the inputs sit in different regions, and the high-confidence softmax collapses into a single-class predictor. CLIP uses cosine similarity to class-prompt embeddings in an L2-normalized 256-d projection space, scaled by a learned temperature. Three effects make this geometry more robust:
  - Cosine similarity is bounded in $[-1, 1]$ — no unbounded margin for the model to over-extrapolate.
  - The L2-normalized space concentrates features near the unit sphere; samples that drift in absolute feature space remain closer in *angular* terms to the prompt anchors.
  - The learned temperature τ acts as a soft regularizer on the cosine distribution: it tempers extreme confidence without removing the ranking signal.
- This is a property of the CLIP *paradigm*, not of any specific encoder. The same geometry was designed for the open-vocabulary regime (new class prompts at inference time) and incidentally protects against distribution shift on closed-set tasks. **This validates RQ2's foundational-model framing**: CLIP earns its complexity not at peak in-distribution accuracy, but in the open-vocabulary capabilities (retrieval, prompt-tuning at inference) and the robustness-under-shift behaviour shown here.

4.6 Text encoder pretraining

4.6.1 Configuration

- Same arch / loss / prompts / split as the baseline; only the frozen text encoder changes.
- Three encoders: `bert-base-uncased` (the baseline's text encoder), BioBERT, PubMedBERT.
- Two prompt strategies: **name_only** (just the class name, e.g. `‘‘benign lung tissue’’`) and **composed** (`‘‘A histopathology image of {class}.’’`, the baseline format).
- $3 \times 2 = 6$ cells. All three encoders are BERT-family, all produce 768-d hidden state, all tokenize through BERT-family WordPiece tokenizers.
- Tests RQ4 (does biomedical text pretraining help under short class-name prompts?).

4.6.2 NCT-CRC grid (6 cells)

Table 4.6 6-cell biomedical text encoder ablation, macro-F1 on NCT-CRC. **None of the 6 cells beats the baseline** (0.866 **on NCT).** LC25000 in-distribution saturates uniformly; all generalization numbers land in the 0.81–0.86 band. The BERT + composed cell is essentially baseline (same architecture, same prompt template); the best *biomedical* cell — PubMedBERT + name_only at 0.855 — still trails baseline by −0.011 F1..

Cell	**LC25000**	**NCT-CRC**	**Δ vs baseline NCT**
BERT + name_only	0.999	0.821	−0.045
BERT + composed	0.999	0.863	−0.003
BioBERT + name_only	0.998	0.840	−0.026
BioBERT + composed	0.998	0.818	−0.048
PubMedBERT + name_only	0.995	0.855	−0.011
PubMedBERT + composed	0.996	0.808	−0.058

- LC25000 in-distribution: all cells saturate uniformly. The training-source axis does not discriminate these encoders.
- NCT generalization: all 6 cells land in the 0.81–0.86 band. None beats the baseline.

4.6.3 Generalization to Chaoyang and LungHist700

- The best biomedical cell on NCT — PubMedBERT + name_only — was extended to Chaoyang and LungHist700 to test whether biomedical text encoders carry any advantage under heavier shift:

Table 4.7 Best biomedical text-encoder cell (PubMedBERT + name_only) vs baseline on the three external evaluation datasets. The biomedical text encoder ties baseline on Chaoyang (within noise) and regresses on both NCT and LungHist700. No measurable transfer benefit on any of the three datasets..

External dataset	**Baseline F1**	**PubMedBERT+name_only F1**	**Delta**
NCT-CRC-HE-7K (close-shift colon)	0.866	0.855	−0.011
Chaoyang (different-hospital colon)	0.786	0.795	+0.009
LungHist700 (different-source lung, 20×)	0.305	0.279	−0.026

- **No transfer.** The best biomedical cell ties baseline on Chaoyang (+0.009, within noise) and *regresses* on LungHist700 (−0.026). Combined with the NCT result (−0.011), no biomedical-text-encoder configuration tested produces a measurable gain on any external dataset.
- **LungHist700 collapse signature.** Both baseline and PubMedBERT+name_only collapse on lung, just to different classes. Baseline predicts “lung squamous cell carcinoma” on most inputs in the degenerate

regime; PubMedBERT+name_only also collapses to lung_scc but harder ($\sim 92\%$ of predictions). In both cases the per-class recall on the other two lung classes is near-zero — a single-class predictor regime where the text encoder choice only affects how *sharply* the collapse manifests.

4.6.4 What we learned

- Biomedical text pretraining does *not* help on this task at frozen-backbone scale with short class-name prompts. Confirmed across three external datasets: PubMedBERT + name_only (the best biomedical cell on NCT) ties baseline on Chaoyang and regresses on both NCT and LungHist700.
- **Mechanism — why doesn't "a more domain-aware text encoder" improve results?** Three converging reasons:
  - *The projection head absorbs encoder-specific differences.* BERT, BioBERT, and PubMedBERT all start from different 768-d embedding spaces, but they all get pulled through the same projection head trained on LC25000. The head learns to map any reasonable starting embedding to useful class anchors. With only 5 LC25000 classes to separate, this is an easy adaptation regardless of where the starting embeddings sit.
  - *Short class-name prompts don't expose biomedical-vocabulary richness.* Biomedical pretraining adds value most when domain vocabulary actually appears — clinical notes, paper abstracts, multi-sentence text. Strings like `''colon adenocarcinoma''` tokenize into 2–3 wordpieces that any BERT-family encoder handles competently. There is no rare-medical-term upside for biomedical pretraining to surface.
  - *CLIP's shift robustness is geometric, not domain-knowledge driven.* As argued in Section 4.5, CLIP's edge under shift comes from the cosine + L2-norm + temperature classifier geometry, which is independent of the text encoder. Swapping in a more medically-aware text encoder doesn't change the classifier geometry; it changes the starting position of the prompt anchors, which the projection head then re-maps anyway.
- **Small but interesting secondary observation**: vanilla BERT prefers the longer **composed** prompt format; biomedical encoders prefer **bare class names**. Plausible mechanism: biomedical pretraining made strings like `''colon adenocarcinoma''` carry strong signal on their own, while adding generic boilerplate (`''A histopathology image of ...''`) dilutes the prompt embedding for biomedical encoders. Not enough to flip any cell to winning — but a useful methodological note for short-prompt design.

4.7 Image encoder pretraining

4.7.1 Configuration

- Same arch / loss / prompts / split / text encoder (frozen BERT base) as the baseline. The only axis that changes is the image encoder's pretraining.
- Three image encoders compared, all frozen, trainable parameters limited to the projection head + temperature:
  - **ResNet50 (general-purpose, ImageNet pretraining)** — the baseline.
  - **Visual Transformer (ViT-B/32) with PLIP's OpenPath pretraining** — a ViT pretrained on $\sim$208k histopathology image–text pairs scraped from Twitter.
  - **Visual Transformer (ViT-B/14) with DINOv2 self-supervised pretraining** — a ViT pretrained on $\sim$142M curated general natural images via self-distillation, no labels, no histopathology data.
- **Role of DINOv2 in the comparison**: a control. PLIP differs from the baseline on multiple axes simultaneously (architecture: ViT vs CNN; objective: contrastive vs supervised; pretraining domain: histopathology vs natural images). DINOv2 isolates the "ViT + self-supervised on general data" axis from the "histopathology pretraining specifically" axis. If DINOv2 also breaks the baseline's lung

collapse, the win is partly architectural; if it does not, the win is specifically about histopathology pretraining.

- **Caveat**: the two ViTs differ in patch size (PLIP B/32, DINOv2 B/14). At 224×224 input this gives different feature granularity (49 vs 256 patch tokens). We discuss this as a confound in Methodology and address it as a limitation in Chapter 5.

4.7.2 Training source results

- PLIP image encoder + LC25000 supervision: LC25000 macro-F1 = 0.996. Same saturation null as every other trained variant.
- DINOv2 image encoder + LC25000 supervision: LC25000 macro-F1 = 0.989. The saturation null holds even though DINOv2 has never seen labelled histology — its pretraining is self-supervised on general natural images, with no class labels. Confirms that the in-distribution null is backbone-independent: the bottleneck on the meaningful axis is not what the image encoder "knows about histology" in the supervised sense; it is what kind of variation the encoder learned to be sensitive to.

4.7.3 Generalization to external datasets

Table 4.8 Image encoder pretraining swap, macro-F1 across the training-source split and three external datasets. Everything except the image encoder is identical across rows..

Image encoder	LC25000	NCT	Chaoyang	LungHist700
ResNet50 (ImageNet)	0.996	0.866	0.786	0.305
ViT-B/14 (DINOv2 / self-supervised)	0.989	0.855	0.776	0.457
ViT-B/32 (PLIP / OpenPath)	**0.996**	**0.933**	**0.813**	**0.640**

- **NCT-CRC (close-shift colon)**: PLIP wins by +0.067 F1 over the ResNet50 baseline. DINOv2 lands at −0.011 vs baseline — general-domain self-supervision does *not* help on the easiest external dataset. The colon ceiling around 0.88 that none of the other interventions could clear is broken specifically by changing the image encoder's pretraining *domain*, not by architecture or self-supervision alone.
- **Chaoyang (different-hospital colon)**: PLIP wins by +0.027 F1; DINOv2 lands at −0.010 vs baseline. Same colon-side pattern as NCT: only histopathology pretraining helps; general SSL is interchangeable with ImageNet supervision.
- **LungHist700 (different-source lung)**: this is where the control pays off. PLIP wins by +0.335 F1 over baseline; DINOv2 wins by +0.152 F1. *Roughly half of PLIP's lung advantage comes from architecture + self-supervised pretraining on general images; the other half comes from histopathology pretraining specifically.* DINOv2 partially rescues the lung collapse (per-class recall 0.46/0.52/0.40, balanced) but does not match PLIP's full functional classifier (0.51/0.49/0.91).

Per-class signature on LungHist700. Both ResNet50 variants converge to the same degenerate predictor on lung tissue: baseline recall 0.19/0.07/0.89 for normal/AdC/SqC, plain CE 0.15/0.05/0.91 — both predict "squamous" on $\sim$ 90% of inputs regardless of true class. **DINOv2-encoded variant:** 0.46/0.52/0.40 — balanced across all 3 classes, no degenerate collapse, but lower peak per-class scores than PLIP. **PLIP-encoded variant:** 0.51/0.49/0.91 — balanced *and* retains the high squamous recall, the only variant that does both.

4.7.4 What we learned

- The **image encoder's pretraining is the dominant lever in this study**, but the DINOv2 control reveals that "pretraining" decomposes into two contributing components.
- Effect-size summary: PLIP gains over baseline by +0.067 F1 on close-shift colon, +0.027 on different-hospital colon, +0.335 **on different-source lung**. DINOv2 gains over baseline by −0.011 / −0.010 / +0.152 on the same three datasets.

- **On colon (NCT, Chaoyang)**: only histopathology pretraining moves the needle. General-domain self-supervised pretraining (DINOv2) lands within noise of ResNet50 supervised pretraining on natural images. The colon advantage is specifically about *what* the encoder was pretrained on, not *how*.
- **On lung (LungHist700)**: the DINOv2 and PLIP-zero-shot controls together yield a clean three-step decomposition of PLIP-backbone's $+0.335$ F1 gap. Baseline at 0.305 $\rightarrow$ DINOv2 at 0.457 ($+0.152$, "ViT $+$ general-domain SSL" contribution) $\rightarrow$ PLIP zero-shot at 0.544 ($+0.087$, "histopathology pretraining without LC25000 supervision" contribution) $\rightarrow$ PLIP-backbone at 0.640 ($+0.096$, "LC25000 supervision on top of histopathology pretraining" contribution). Three roughly comparable steps of ~ 0.1 F1 each, sequentially adding architecture/SSL, then domain, then task supervision.
- Why colon and lung behave differently, on the ResNet50 baseline: ImageNet supervision contains no histology. LC25000 supervision rescues ResNet50 features that happen to correlate with colon class labels in the LC25000 training distribution. That correlation transfers partially to NCT/Chaoyang (both colon, stylistically close). It does not transfer to lung tissue at all — the lung-side "success" in-distribution is memorisation on the LC25000 lung subset, not learned lung tissue features.
- Why DINOv2 helps on lung but not colon: DINOv2's self-supervised pretraining on $\sim$ 142M diverse natural images produces image features that are sensitive to a much broader range of local-texture variation than supervised ImageNet pretraining. That broader feature space provides *enough* signal for LC25000 supervision + the projection head to construct a usable lung classifier, even though the encoder has never seen histology. On colon, where ResNet50 features were already enough for partial transfer, DINOv2's broader features do not help further — the bottleneck on colon is the pretraining domain, not the architecture's feature breadth.
- Patch-size confound check: DINOv2 has more spatial detail than PLIP (256 vs 49 patch tokens at the same 224×224 input), but still loses on colon. The patch-size difference is therefore not the dominant factor here — pretraining is.

4.8 Discussion

4.8.1 Synthesis across experiments

- This chapter evaluated six interventions on a frozen general-purpose CLIP pipeline (ResNet50 + BERT). Five of the six (stain normalization, biomedical text encoders, plain-classifier substitution, the prompt-format variations within the text-encoder ablation) produce changes within ~ 0.03 F1 of the baseline on close-shift external evaluation (NCT) and at most ~ 0.03 F1 of improvement on the harder external datasets. Stain Macenko is the only input-side lever that moves the needle positively under shift ($+0.030$ on Chaoyang, $+0.020$ on LungHist700), but its magnitude is dominated by the image-encoder lever on the same datasets. The best biomedical text-encoder cell (PubMedBERT + name_only) confirms the null result across all 3 external datasets — ties baseline on Chaoyang, regresses on NCT and LungHist700.
- The sixth intervention — replacing the image encoder's pretraining (Section 4.7) — moves the headline metric by $+0.067$ on close-shift colon (NCT), $+0.027$ on different-hospital colon (Chaoyang), and $+0.335$ **on different-source lung (LungHist700)**, taking the lung result from a degenerate single-class predictor at 0.305 to a functional 3-class classifier at 0.640.
- The DINOv2 and PLIP-zero-shot controls together decompose that gain on lung into a clean ladder: 0.305 (baseline) $\rightarrow$ 0.457 (DINOv2: ViT + SSL contribution, $+0.152$) $\rightarrow$ 0.544 (PLIP zero-shot: histopath pretraining contribution, $+0.087$) $\rightarrow$ 0.640 (PLIP-backbone: LC25000 supervision on top, $+0.096$). Three roughly equal steps of ~ 0.1 F1. On colon, the decomposition collapses — DINOv2 ties baseline and PLIP zero-shot actually falls below baseline on Chaoyang (0.731 vs 0.786) because LC25000 supervision was already enough.
- **The image encoder's pretraining is the dominant lever**, by roughly $5\times$ over the next-largest single lever on colon external evaluation. On lung, no head/loss/text-side intervention rescues a non-functional predictor at all; only changing the encoder does.

- The contemporary methods literature's emphasis on novel loss functions, text-side innovations, and stain harmonisation is not supported by this study at the frozen-backbone scale we evaluated.

4.8.2 Secondary observations

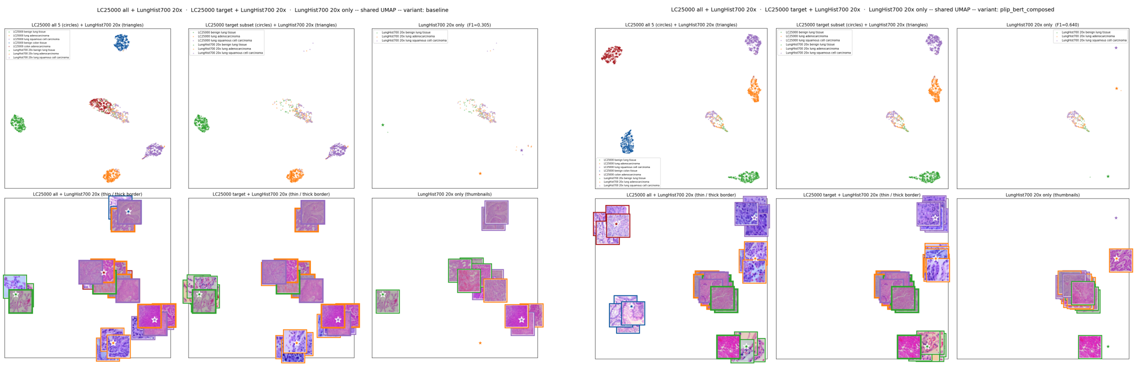
- **Cross-organ regime change.** ResNet50 variants partially work on colon external data (0.78–0.88) but *categorically fail* on lung (~ 0.30, near 3-class chance). Mechanism: LC25000 supervision can rescue ResNet50 features that happen to correlate with colon class labels in LC25000, but cannot produce useful lung-tissue features within a frozen ImageNet ResNet50; the in-distribution lung "success" is memorisation. Per-organ external evaluation is therefore essential; aggregating across datasets hides the regime change.
- **Classification quality $\neq$ representation alignment.** Baseline CLIP achieves cross-dataset *cluster alignment* but lower classification F1; PLIP achieves the highest F1 but leaves external samples in their own region of feature space. The contrastive objective optimises same-class-pair alignment; the cosine-classifier rule only needs alignment to the class-prompt direction. These are different geometric properties — the right tradeoff depends on whether the downstream task is classification or representation-driven (retrieval, clustering, transfer).
- **Plain-classifier flip under shift.** Plain CE ties CLIP in-distribution and marginally wins NCT, but loses Chaoyang ($+0.064$ for CLIP) and LungHist700 ($+0.024$). High-confidence softmax classifiers degenerate into single-class predictors as shift severity grows (visible in the Chaoyang per-class signature); CLIP's cosine geometry doesn't. This supports the foundational-model framing of RQ2: CLIP earns its complexity in shift robustness, not in-distribution accuracy.

4.8.3 Headline summary

Table 4.9 Headline summary across the chapter. PLIP-pretrained ViT wins every external evaluation; the lung-side advantage ($+0.335$ F1 over baseline) is an order of magnitude larger than the colon-side advantages. DINOv2 (the same-architecture, general-domain self-supervised control) lands between baseline and PLIP on lung but ties baseline on colon, decomposing PLIP's lung advantage into roughly equal architecture/SSL and histopathology-pretraining contributions. Stain Macenko (NCT-NORM) helps on the two datasets that arrive in their native stain palette (Chaoyang $+0.030$, LungHist700 $+0.020$) and regresses on NCT (-0.022), which arrives already Macenko-normalised at release. The best biomedical text-encoder cell (PubMedBERT + name_only) ties baseline on Chaoyang and regresses on both NCT and LungHist700 — no measurable transfer benefit on any of the three datasets. Limitations and future work are discussed in Chapter 5..

Variant	LC25000	NCT	Chaoyang	LungHist700
ViT-B/32 (PLIP)	**0.996**	**0.933**	**0.813**	**0.640**
ViT-B/14 (DINOv2)	0.989	0.855	0.776	0.457
Baseline (ResNet50 + BERT)	0.996	0.866	0.786	0.305
Plain CE classifier	0.995	0.880	0.722	0.281
Stain Macenko (NCT-NORM reference)	0.986	0.844	0.816	0.325
Best biomedical text-encoder cell (PubMedBERT + name_only)	0.995	0.855	0.795	0.279
PLIP zero-shot (no LC25000 supervision)	0.700	0.661	0.731	0.544

Figure 4.8 6-cell UMAP grid (one panel per BERT/BioBERT/PubMedBERT × name_only/composed cell), cross-distribution view on NCT-CRC. The clusters are qualitatively similar across cells, consistent with the lack of meaningful F1 separation in Table 4.6..



(a) ResNet50 baseline: a single tight blob, no internal class structure.

(b) ViT-B/32 with PLIP pretraining: three visibly separated sub-clusters by class.

Figure 4.9 LungHist700 cross-distribution UMAP comparison — the visual headline of this chapter. The ResNet50 baseline (a) places all 359 LungHist700 samples into a single tight cluster with no internal class structure; the PLIP-pretrained ViT (b) produces three visibly distinct sub-clusters by class, sitting adjacent to (but not co-located with) the matching LC25000 lung clusters..

5 Conclusions and Future Work

5.1 Conclusions

Headline. The image encoder's pretraining is the dominant lever in this study. Holding everything else fixed, swapping the general-purpose ResNet50 (ImageNet supervised) for PLIP's pretrained ViT-B/32 (histopathology image–text contrastive) improves external evaluation by $+0.067$ F1 on close-shift colon (NCT-CRC), $+0.027$ on different-hospital colon (Chaoyang), and $+0.335$ on different-source lung (LungHist700). None of the other five interventions tested moves the needle comparably. The DINOv2 control reveals that roughly half of PLIP's lung advantage comes from architecture + self-supervised pretraining alone, the other half from histopathology pretraining specifically.

RQ1 — Baseline behavior and failure modes

- The frozen general-purpose CLIP pipeline (ResNet50 $+$ BERT) saturates LC25000 at ~ 0.99 macro-F1, transfers partially to colon external data (0.866 NCT, 0.786 Chaoyang), and categorically fails on lung external data (0.305 on LungHist700, near 3-class chance).
- The lung failure is a regime change, not a smooth degradation: both ResNet50-based variants collapse into a single-class predictor (predicting "squamous" on $\sim 90\%$ of inputs regardless of true class).
- See Results §4.2.

RQ2 — CLIP's foundational-model premise

- Validated under shift, not in-distribution. CLIP and a plain CE classifier on the same frozen backbone tie at saturation in-distribution and tie marginally on close-shift external (NCT, plain CE $+0.014$); CLIP's edge appears under heavier shift (Chaoyang $+0.064$ for CLIP, LungHist700 $+0.024$ for CLIP).
- Mechanism: CLIP's bounded-cosine $+$ L2-normalised $+$ temperature-scaled classifier geometry is more robust than the high-confidence softmax, which degenerates under shift into a single-class predictor.
- See Results §4.5.

RQ3 — Stain normalization and generalization

- Direction is data-dependent. Macenko(NCT-NORM) at training, with the same transform applied at inference only on OOD datasets that arrive in their native stain palette, *helps* on Chaoyang ($+0.030$ F1) and LungHist700 ($+0.020$ F1) and *regresses* on NCT-CRC (-0.022 F1, which is already Macenko-normalised at release).
- Mechanism: when an OOD input genuinely sits in a different stain palette than the training distribution, applying Macenko at inference moves it closer to the model's training-stain space and the frozen

backbone produces better features. On NCT, where the input was already approximately in the target stain space, training-time Macenko added a non-linear input-statistic shift the frozen backbone could not absorb for no gain at evaluation.

- Magnitude is modest. Even where Macenko helps, the gain is comparable to the image-encoder lever on Chaoyang ($+0.027$ for PLIP) and an order of magnitude smaller on LungHist700 ($+0.335$ for PLIP). Stain normalization is a useful but not dominant lever.
- See Results §4.3.

RQ4 — Biomedical text encoder pretraining

- No effect on the meaningful axis. None of the 6 cells (BERT / BioBERT / PubMedBERT $\times$ bare-name / composed-prompt) beats the baseline on NCT external evaluation; all cells fall within 0.81–0.86 macro-F1.
- Extended to 3 external datasets via the best biomedical cell (PubMedBERT + name_only): no transfer benefit. Ties baseline on Chaoyang ($+0.009$, within noise), regresses on NCT (-0.011) and LungHist700 (-0.026). On LungHist700 the model collapses to a single-class predictor ($\sim 92\%$ squamous), same regime as baseline but with sharper collapse.
- Mechanism: the trained projection head absorbs encoder-specific differences; short class-name prompts don't expose biomedical vocabulary richness. CLIP's shift robustness is a property of the classifier geometry, not of the text encoder's domain knowledge.
- Secondary observation: BERT prefers the longer composed prompt; biomedical encoders prefer bare class names — small but useful for short-prompt design.
- See Results §4.6.

RQ5 — Image encoder pretraining (the dominant lever)

- Replacing ResNet50 with PLIP's pretrained ViT-B/32 (histopathology image–text contrastive) gains $+0.067$ F1 on NCT, $+0.027$ on Chaoyang, and $+0.335$ on LungHist700. None of the head/loss/text-side interventions tested gets close.
- The DINOv2 control (ViT-B/14 with general-domain self-supervised pretraining, no histopathology data) decomposes the lung gain: DINOv2 alone lifts LungHist700 by $+0.152$ over baseline, indicating that $\sim$half of PLIP's lung advantage comes from "ViT + self-supervised pretraining on diverse natural images" and $\sim$half from "histopathology pretraining specifically".
- On colon, only histopathology pretraining helps; general-domain SSL ties baseline within noise. The lever's contribution is dose-dependent on shift severity.
- See Results §4.7.

RQ6 — Generalization across 3 external datasets and 2 organs

- PLIP's ViT-B/32 wins every external evaluation in the study (0.933 NCT, 0.813 Chaoyang, 0.640 LungHist700). The lung-side advantage over baseline ($+0.335$ F1) is an order of magnitude larger than the colon-side advantages ($+0.067$ NCT, $+0.027$ Chaoyang).
- The cross-organ asymmetry is a regime change: ResNet50-based variants transfer partially to colon (0.78–0.88 F1) but categorically fail on lung (~ 0.30, near chance). PLIP shows no such asymmetry.
- Methodological consequence: per-organ external reporting is essential; aggregating across datasets would have hidden the regime change.
- See Results §4.7 and §4.8.

5.2 Limitations

- LC25000 is single-institution; no patient-level grouping or clinical validation. Three external datasets (NCT-CRC, Chaoyang, LungHist700) span shift severity and two organs but each is itself a single-institution release.
- LC25000 augmentation-pair leakage: 1 250 source images $\times$ 20 augmentations are scattered randomly across the 80/10/10 split. Likely the dominant inflater of in-distribution numbers; external evaluations are unaffected.
- 1 280 byte-identical LC25000 duplicates cross train/test under the canonical pre-dedupe split. Effect $\sim$1–2 F1 points on in-distribution numbers; external evaluations unaffected.
- All variants use a frozen image backbone with a trained projection head and temperature. Full end-to-end fine-tuning is out of scope by design; this study tests the frozen-backbone transfer regime, which is the practical setting for histopathology foundation-model downstream use.
- Single seed per condition; seed variance not estimated. Especially load-bearing for the LungHist700 result ($N = 359$ images, per-class supports 85/146/128). Bootstrap confidence intervals on the headline lung gap are listed as a high-priority future-work item.
- External evaluation uses restricted argmax over the LC25000-mappable classes: 2-class colon for NCT and Chaoyang, 3-class lung for LungHist700. Richer multi-class external scenarios are out of scope.
- Patch-size confound in the image-encoder ablation: PLIP (ViT-B/32) and DINOv2 (ViT-B/14) differ in both pretraining domain and patch size (49 vs 256 tokens at 224×224 input). DINOv2's finer patches do not rescue the colon-side gap, indicating patch size is not the dominant factor, but a fully controlled comparison would match patch sizes.
- Stain normalization is evaluated with a single NCT-NORM reference patch for the external evaluation; reference rotation could shift the result.
- LungHist700 source bias is not characterised; the lung-side ResNet50 failure could be partly stain-shift artefact in addition to the architecture limitation.
- No human-expert evaluation; no clinical validation.
- No head-to-head against other histopathology foundation models (UNI, CONCH, Virchow-2, Biomed-CLIP, MUSK, PathChat) — compute and time scope.

5.3 Future Work

Note: this section will be expanded once the rest of the thesis is locked, to ensure the planned follow-ups reflect the final picture of findings.

Medium-term

- Bootstrap confidence intervals on the headline LungHist700 +0.335 F1 gap. $\sim$1 hour analysis on saved predictions, no retraining. Defends the claim against the small-N critique ($N = 359$ images, per-class supports 85/146/128).
- Extend the PLIP zero-shot reference to Chaoyang and LungHist700. Closes the gap so the reference is comparable across all four datasets.
- Larger effective batch (gradient accumulation or larger GPU) to test whether the InfoNCE false-negative regime is binding in our setup.
- Prompt-tuning methods (CoOp [28], CoCoOp) under frozen encoders — can a learned prompt close some of the colon-side gap without changing the backbone?

Longer-term

- Architecture-controlled comparison against other histopathology foundation models: UNI [21], CONCH [20], BiomedCLIP [22], MUSK [23]. Disambiguates “PLIP specifically wins” from “any histopathology-pretrained backbone wins”.
- Multi-institution training and evaluation.
- Returning to segmentation with pathology foundation-model backbones.
- CLIP-with-short-labels applied to other low-label scientific domains (e.g., the tutor’s gravitational-physics work).

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